University of Rajshahi
Dept. Computer Science and Engineering
University of Rajshahi
Semester Final Examination, B.Sc. Engg. 2019, 1st year, Odd semester

Course ID: CSE 1121 Course Title: : Structural Programming Language

Total Time 3 Hours Total Marks 52.5

(Answer any six questions taking three from each section)

1. (a) Why do we get "segmentation fault" message in C programming? 02
   (b) What will be the value of c and d if 03
   int a=10; int b=3; float c,d; c=a/b; d=a/2.0;
   (c) Can the following program sum all the numbers in the array num[ ]? If not, modify only one variable or a value so that you can sum first ten numbers in the array. 3.75

```
#include<stdio.h>
int main()
{
   int num[11]={5,10,15,20,0,30,35,40,45,50,0};
   int i=0,sum=0;
   while(num[i%5])
   {
      sum=sum+num[i];
      i=i+1;
   }
   printf("%3d",sum);
   return 0;
}
```

2. (a) What is recursive function? What is its limitation? 02
   (b) Write a C program for the following problem: 6.75

The student association of the dept. of CSE of Rajshahi University arranged a game in their last picnic. In that game; one student, X from 1st year and another student, Y from 2nd year were asked to pick up balls as many as possible from a basket within 2 minutes. The basket contained many balls having three different colors; red (R), green (G) and blue (B). The winner was decided based on earned points (P), as calculated by the formula; P = (RX-RY)*7 + (GX-GY)*5 + (BX-BY)*3, where RX and RY are the number of red colored balls picked up by X and Y, respectively and so on. The positive value of P indicated that the winner was X; otherwise, the winner was Y.

Input: The input will have two lines; the first line is for X and the second line is for Y. Each line will have three integers separated by spaces; describing the number of red, green and blue balls picked up by the respective student. The inputs will be given in such a way that the value of P will either be positive or negative.

Example:
17 24 30
15 25 25
Output: The output will have a single line that shows the name of the winning year. For example, the output for the above input set will be:
1st year

3. (a) Find out the output of the following program. 4.75

```
#include<stdio.h>
#include<conio.h>

void pointer(int *p, int *q, int *r, int *s);

main() {
     int a,b,c,d,*x,*y;
     a=10; b=20; x=&c; y=&d; c=30; d=40;
     printf("Before calling  %d %d %d %d \n",a,b,*x,*y);
     pointer(&c,&d,&a,&b);
     printf("After calling  %d %d %d %d\n",a,b,*x,*y);
}
```

2023/03/05 11:21

CamScanner

```
tempnum = *num1;
  *num1 = *num2;
  *num2 = tempnum;
}
int main( )
{
int v1 = 11, v2 = 77 ;
printf("Before operation:");
printf("\nValue of v1 is: %d", v1);
printf("\nValue of v2 is: %d", v2);

  operation( &v1, &v2 );

printf("\nAfter operation:");
printf("\nValue of v1 is: %d", v1);
printf("\nValue of v2 is: %d", v2);
}
```

(b) A 4x4 matrix is given in a file named in.txt. **5.75**

in.txt:

10	15	20	25
5	10	20	15
15	25	10	5
20	5	15	10

Read the matrix from the in.txt file, transform all the decimal numbers to its equivalent binary numbers and write into out.txt file.

out.txt:

1010	1111	10100	11001
101	1010	10100	1111
1111	11001	1010	101
10100	101	1111	1010

Section **B**
Answer any **THREE** questions.

5. (a) Some numbers are given as input and their average 214.400000 is calculated using getAverage() function. However, getAverage() is not defined. Your task is to define getAverage() function. **3.00**

```
int main () {
int balance[5] = {1000, 2, 3, 17, 50};
  double avg;
 avg = getAverage( balance, 5 ) ;
printf( "Average value is: %f ", avg );
  return 0;
}
```

(b) A programmer developed the following program to have the expected output as given below. However, expected output is not found. **5.75**

(i) What is the current output?
(ii) What is the error?
(iii) Make necessary correction.

Expected Output:

```
c
b
cb
a
ca
```

2023/03/05 11:21

CamScanner

```
printf("\nEnter the marks of of a student of roll %d \n",s1.x, s1.y);
scanf("%f",&s1.y);
printf("\nRoll number is %d and marks is %f",s1.x,s1.y);

printf("\nUsing Union \n");
printf("Enter the roll of a student\n");
scanf("%d",&s2.x);
printf("\nRoll number is %d and marks is %f",s2.x,s2.y);
printf("\nEnter the marks of of a student of roll %d \n",s2.x, s2.y);
scanf("%f",&s2.y);
printf("\nRoll number is %d and marks is %f",s2.x,s2.y);
}
```

8. (a) Let the contents of a file named "Test.dat" are: 4

1 2 3 4 5
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9

Write a program to read those values from the mentioned file and print the biggest value of each row in another file named "Out.dat".

The contents of "Out.dat" look like:
5 6 7 8 9

(b) What will be the output of the following code? 4.75

```
#include<stdio.h>
#include<conio.h>
int y(int n);

int main(){
int x;
   x=y(10);
printf("Final Output=%d",x);
getch();
}

int y(int n){
printf("The function is called with the argument %d \n",n);
   if(n==0)
     {
printf("In Terminating Condition= %d \n",n);
          return n;
     }
   else
          { printf("In Recursive function is called with =%d\n", (n-2));
          return n+y(n-2);
          }
printf("CSE %d Times\n",n);
}
```


2023/03/05 11:22

CamScanner

University of Rajshahi

Department of Computer Science and Engineering

B.Sc. Engineering **Part-I Odd** Semester Examination **2020**

Course Code: **CSE-1121** Course Title: **Structural Programming Language**

Time: **03** Hours Full Marks: **52.5**

[N.B. Answer 06(Six) questions taking any 03(Three) questions from each section. The figures in the right margin indicate full marks.]

Section-A

1. (a) What is meant by computer program and programming language? Why C is called structured programming language? 3

   (b) What is an identifier? What are the rules for naming an identifier? Which is not a valid C identifier from the following list and why? 3.75
   (i) my name (ii) my_name (iii) struct (iv) stop! (v) void

   (c) Evaluate the following expression. Assume, int a= 10, b=20, c=5, e=2; float d=4.5; 2

   a+2 < b || !c && a==d || a-2 <= e

2. (a) What is run-time error? Give some examples of run-time error? 2.00

   (b) List the syntax error (if any) of each line of the following code 4.00

```
#includ<stdio>
void function(int y)
{
y=y*y;
return ;
}
main()
{
Intx,y;
scanf("%dd",&x,&y);
switch (x)
   {
    case(x<0): printf("x is positive"); break;
    case(x>0); printf("x is negative");
   }
x=function(y);
printf("Final value of = %d\n", x);
}
```


2023/03/05 11:20

CamScanner

Section-B

5. a) Define pointer. What are the differences between malloc () and calloc (). 3.5
   b) What will be the output of the following program when executed?

```
void func (int, int []);
void main () {
        int arr [] = {1, 2, 3, 4};
        int i;
        func (4, arr);
        for (i = 0; i<4; i++)
            printf ("%d", arr[i]); }
        void func (int n, int arr []) {
        int * p = 0; int i = 0;
        while (i++<n)
          p = & arr [i];
          * p = 0;
        }
```

   c) Distinguish between function call– by–value and call–by–reference with appropriate example. 3.5

6. a) Define recursion. What conditions must be satisfied for a recursive function? 3
   b) Write a 'c' program using recursion to take n as input. Then find out the value of n! using recursive function. 3.5
   c) What will be the output of the following programs when executed? 3.5

```
i) void main () {
   int a = 500, b = 100, c;
   if (!a>=400) b=300; c=200;
        printf ("b=%d  c=%d\n", b, c); }
ii) void main () {
   int i=3, *j, k;
   j = &i;
   printf ("%d\n", i* *j*i+*j); }
```

7. a) What are the purposes of using Global and external variables? Describe with examples. 3
   b) Write a complete program to sort a set of integer numbers in ascending order. 3.5
   c) Write a c program that will exchange value of two variables without using third or temporary variable. 3.5

8. a) Define structures. How does the structure variable declaration differ from structure type declaration? 3
   b) Compare union and structures. When union is more useful than structure? Explain with example. 2.5
   c) Define file pointer. Suppose 200 characters are stored in a file named "data.txt". In each line of the file 20 characters are stored. All the characters are separated by single space. Write a complete program to open the file and determine the frequency of each character. Finally, write the frequency in another file named "frequency.txt". 4.5

2023/03/05 11:20

CamScanner

```
void pointer(int *p, int *q, int *r, int *s)
{
    *p=100; *q=200; *r=300; *s=400;
}
```

(b) Find out the output of the following program:

```
#include<stdio.h>
void pointer(int *point)
{
   int i;
   for (i=0;i<10;i++)
   {
      printf("%3d",*point);
      point++;
   }
}
int main()
{
   int i;
   int num[10]={0,1,2,3,4,5,6,7,8,9};
   pointer(num);
   printf("\n");
   for(i=0;i<10;i++)
   printf("%3d",num[i]);
   return 0;
}
```

4. (a) Write a program to toggle a bit of a number at nth position using bit-wise operator? 03
   (b) Explain 'the name of array can be considered as pointer'. Give an example how you can pass an array to a function. 03
   (c) Can the following program sum the three numbers? If not, why? 2.75

```
#include<stdio.h>

int main()
{
   int a,b,c,sum=0;
   if(scanf("%d%d%d",&a,&b,&c)==3)
   {
      sum=a+b+c;
   }
   printf("%3d",sum);
   return 0; }
```

Section **B**

Answer any **THREE** questions.

5. (a) Write a program to display A to Z and a to z using ASCII code. 4.75
   (b) Sample input and a program are given. Find out the output of the following program. 04

Sample Input: Happy Coding!

```
#include<stdio.h>
#include<string.h>
void func(char str[])
{
   for (int i=0; str[i]!='\0'; i++)
   {
      if (str[i]>='A' && str[i]<='Z')
         str[i] = str[i] + 'a' - 'A';
      else if (str[i]>='a' && str[i]<='z')
         str[i] = str[i] + 'A' - 'a';
   }
}
```

2023/03/05 11:21

CamScanner

```
int c = 0;
int d=0;

printf ("value of a in main() = % \n",  a);
 c = sum( a, b);
 d = sub( a, b);
printf ("value of c in main() = %d\n",  c);
printf ("value of d in main() = %d\n",  d);
printf ("value of e in main() = %d\n",  e);
 return 0;
}

int sum(int a, int b) {
printf ("value of a in sum() = %d\n",  a);
printf ("value of b in sum() = %d\n",  b);
  return a + b;
}

int sub(int a, int b) {
int c=40;
printf ("value of a in sub() = %d\n",  a);
printf ("value of b in sub() = %d\n",  b);
  return a-b + c;
}
```

(b) What will be the output of the following code? Is there any better or easier way to produce the same output? If any, write the program? **5**

```
int main()
   {
int v[4][4];
int n=3;
     for(int i = 1;i <=n;++i)
     for(int j = 1;j <=n;++j)
      v[i-1][j-1]=(i/j)*(j/i);

      for(int i = 1;i <=n;++i)
      {
         for(int j = 1;j <=n;++j)
     {
printf("%d",v[i-1][j-1]);
     }
printf("\n");
      }

return 0;
  }
```

3. (a) How do logical errors differ from syntactic and execution errors? **2.00**
   (b) What are the purposes of using continue and break statement? Give example **3.00**
   (c) What is the purpose of using do-while statement? How does it differ from the while statement? Explain with example. **3.75**

4. (a) What will be the output of the following code? **3.00**

```
void operation(int *num1, int *num2)
{
int tempnum;
```

2023/03/05 11:21

CamScanner

```
int c = 0;
int d=0;

printf ("value of a in main() = % \n",  a);
c = sum( a, b);
d = sub( a, b);
printf ("value of c in main() = %d\n",  c);
printf ("value of d in main() = %d\n",  d);
printf ("value of e in main() = %d\n",  e);
return 0;
}

int sum(int a, int b) {
  printf ("value of a in sum() = %d\n",  a);
  printf ("value of b in sum() = %d\n",  b);
  return a + b;
}

int sub(int a, int b) {
  int c=40;
  printf ("value of a in sub() = %d\n",  a);
  printf ("value of b in sub() = %d\n",  b);
  return a-b + c;
}
```

(b) What will be the output of the following code? Is there any better or easier way to produce the same output? If any, write the program? **5.00**

```
int main()
{
   int v[4][4];
   int n=3;
  for(int i = 1;i <=n;++i)
  for(int j = 1;j <=n;++j)
   v[i-1][j-1]=(i/j)*(j/i);

    for(int i = 1;i <=n;++i)
    {
       for(int j = 1;j <=n;++j)
   {
     printf("%d",v[i-1][j-1]);
   }
    printf("\n");
     }

return 0;
  }
```

3. (a) How do logical errors differ from syntactic and execution errors? **2.00**
   (b) What are the purposes of using continue and break statement? Give example **3.00**
   (c) What is the purpose of using do-while statement? How does it differ from the while statement? Explain with example. **3.75**

4. (a) What will be the output of the following code? **3.00**

```
void operation(int *num1, int *num2)
{
  int tempnum;
```

2023/03/05 11:22

CamScanner

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering **Part-I, Odd** Semester Examination **2018**
Course Code: **CSE-1121** Course Title: **Structural Programming Language**
Time: **03** Hours Full Marks: **52.5**

Section **A**
Answer any **THREE** questions.

1. (a) What is run-time error? Give some examples of run-time error? **2.25**
   (b) If i=4, j=2.5 and int answer, what will be the output of answer=(i+j)%2? **1.50**
   (c) List the syntax error (if any) of each line of the following code. **5.00**

```
struct Books {
   char  title[50];
   char  author[50];
   char  subject[100];
int book_id;
};
double printBook( struct Books book ) {
printf( "Book title : %f\n", book.title);
printf( "Book author : %s\n", book.author);
printf( "Book subject : %s\n", book1.subject);
printf( "Book book_id : %s\n", book.book_id);
}

int main( ) {

struct Books Book1;
struct Books Book2;

strcpy( Book1.title, "C Programming");
strcpy( Book1--author, "Prata");
strcpy( Book1.subject, "C Programming Tutorial");
  Book1.book_id = '6495407';

strcpy( "Telecom Billing", Book2);
strcpy( Book2.author, "Goodman");
strcpy( Book2.subject, "Telecom Billing Tutorial");
  Book2.book_id = "6495700";

printBook( Books Book1 );

printBook( Book2 );

  return 1;
}
```

2. (a) What will be the output of the following code? **3.75**

```
#include <stdio.h>

int a = 20;
int e;

int main () {
int a = 10;
int b = 20;
```

2023/03/05 11:21

CamScanner

to deal with it. In this simplified problem, let us just consider that Google assigns ev
web page an integer-valued relevance.

The most related page will be chosen. If there is a tie, all the pages with the highest relevance are possible to be chosen. Your task is simple, given 10 web pages and their relevance. Just pick out all the possible candidates which will be served to the user when 'I'm feeling lucky'.

Input: The input contains multiple test cases. The number of test cases T is in the first line. For each test case, there are 10 lines, describing the webpage and the relevance. Each line contains a character string without any blank characters denoting the URL of this webpage and an integer Vi denoting the relevance of this web page. The length of the URL is between 1 and 100 inclusively. (1 < Vi < 100)

Output: For each test case, output several lines which are the URLs of the web pages which are possible to be chosen. The order of the URLs is the same as the input. Please look at the sample output for further information of output format.

Sample Input
2
www.youtube.com 1
www.google.com 2
www.google.com.hk 3
www.alibaba.com 10
www.taobao.com 5
www.bad.com 10
www.good.com 7
www.fudan.edu.cn 8
www.university.edu.cn 9
acm.university.edu.cn 10
www.youtube.com 1
www.google.com 2
www.google.com.hk 3
www.alibaba.com 11
www.taobao.com 5
www.bad.com 10
www.good.com 7
www.fudan.edu.cn 8
acm.university.edu.cn 9
acm.university.edu.cn 10
Sample Output
Case #1:
www.alibaba.com
www.bad.com
acm.university.edu.cn
Case #2:
www.alibaba.com

2023/03/05 11:21

CamScanner

(c) Can the following program sum all the numbers in the array num[ ]? If not, modify only one variable or a value in the program so that you can sum first ten numbers in the array. However, you can't update the array. 2.75

```
#include<stdio.h>
int main()
{
int num[13]={5,10,15,20,11,30,35,40,45,50,0,65,20};
int i=0,sum=0;
int j=0;
   for (j=0;num[j%5];j++)
   {
      sum=sum+num[i];
      i=i+1;
   }
printf("%3d",sum);
   return 0;
}
```

3. (a) What will be the output of the following code? 4.25

```
#include <stdio.h>
int a = 20;
int sum(int a, int b)
{
printf ("value of a in sum() = %d\n",  a);
printf ("value of b in sum() = %d\n",  b);
  return a + b;
}
int main () {
int a = 10;
int b = 20;
int c = 0;
printf ("value of a in main() = %d\n",  a);
 c = sum( a, b);
printf ("value of c in main() = %d\n",  c);
 return 0;
}
```

(b) What will be the output of the following code? 4.50

```
#include<stdio.h>
int a,b,c;
void addint(int a)
{
   a=50;
   b=200;
   c=300;
}
```


2023/03/05 11:20

CamScanner

6. (a) Given an array of N numbers and a number K. Write a program to insert a number in the given array such that the bitwise XOR of all the elements in the new array equals the given input K. 4.50

(b) Given an array where every element occurs three times, except one element which occurs only once. Write a program to find the element that occurs once using bitwise operators. 4.25

Sample Input: {12, 1, 12, 3, 12, 1, 1, 2, 3, 3}
Sample Output: 2

7. (a) Define recursive function with example. What are the rules that a recursive function must have to satisfy? 2.5

(b) Distinguish between function call by value and function call by reference. Explain with example. 2.5

(c) Write a recursive function in C programming that will receive an integer n as input and return the sum of the digits of the number. 3.75

Sample Input: 1234
Sample output: 10

8. (a) 4.25

```
vector<int> v;
v.assign(5, 10);
cout<< "The vector elements are: ";
for (int i = 0; i <v.size(); i++)
        cout<< v[i] << " ";
v.push_back(15);
int n = v.size();
cout<< "\nThe last element is: " << v[n - 1];
v.pop_back();
cout<< "\nThe vector elements are: ";
        for (int i = 0; i <v.size(); i++)
                cout<< v[i] << " ";
```

(b) Write a program to display **A** to **Z** and **a** to **z** using ASCII code. 4.50

2023/03/05 11:21

CamScanner

04

```
int main()
{
   char str[20];
   gets(str);
   func(str);
   printf("\n%s",str);
   return 0;
}
```

6. (a) What is header file? 02
   (b) 100 small letters are written in a file named input.txt. In each line, 10 letters are written. All the letters are separated by a single space. Determine the frequency of the letters and write into another file named output.txt. 4.75
   (c) What are the differences between compiler and interpreter? 02
7. (a) How can you optimize a program? Give an example. 03
   (b) How can you extend memory allocation during execution of a program? 2.75
   (c) What will be output of the following program? 3.75

```
#include<stdio.h>
#include<conio.h>
int x[5][5]={   {1, 4, 3, 6, 8},
                {2, 9, 0, 5, 7},
                {5, 9, 6, 7, 6},
                {9, 0, 2, 6, 8},
                {3, 6, 0, 1, 7} };

int i,j,k,l, tmp,big,p;

main() {

 for (i=0;i<=4;i++)
 {
   for(j=0; j<=4; j++)
   {
     for(k=j; k<=4; k++)
     {
       for(l=k ; l<=4; l++)
       {
       x[k][l]=x[k][l]+1;
       }
     }
   }
 }

for (i=0;i<=4;i++)
 {
   for(j=0; j<=4; j++)
   {
     printf ("%d ",x[i][j]);
   }
   printf ("\n ");
 }
getch(); }
```

8. (a) What is the difference between an array and a structure? How can you determine the number of bytes allocated for an array and a structure? 2.75
   (b) Google is one of the most famous Internet search engines which hosts and develops a number of Internet based services and products. In the website, an interesting button 'I'm feeling lucky' attracts our eyes. This feature could allow the user skip the search result page and go directly to the first ranked page. Amazing! It saves a lot of time. 06

   The question is, when one types some keywords and presses 'I'm feeling lucky' button, which web page will appear? Google does a lot and comes up with excellent approaches

2023/03/05 11:21

CamScanner

Section B

(b) What is the header file? 1.5

(b) What is the benefitof 'calling a function by reference'? 1.5

(c) What will be the output of the following code? If you think any output (value)to be garbage, mention it as garbage too. Let the addresses of ***a*** and***x***are 22FF44and 22FF34 respectively. 5.75

```
#include<stdio.h>
#include<conio.h>

main() {
inta,b,c,d,*x,*y;

    a=15;
    x=&a;

printf("a=%x &a=%x x=%x *x=%x\n",a,&a,x,*x);
printf("&x=%x *(&x)=%x \n",&x,*(&x));
printf("&a=%x *(&a)=%x \n",&a,*(&a));

printf("&(*(&a))=%x *(&(*(&a)))=%x \n", &(*(&a)), *(&(*(&a))));
printf("&(*(&x))=%x *(&(*(&x)))=%x \n", &(*(&x)), *(&(*(&x))));

getch();
}
```

6. (a) What is function? Why is it necessary in programming? 2

(b) What is a function prototype? What do you mean by recursion? State the necessary condition of recursive function. 3

(c) Write a C function to evaluate the series: 3.75

$$\sin(x) = x - (x^3/3!) + (x^5/5!) - (x^7/7!).....$$

7. (a) What are the differences between 'structure' and 'array'? 2

(b) What is run-time error? Give some examples of run-time error? 2

(c) What will be the output of the following code? If you think any output (value) to be garbage, mention it as garbage too, consider the input for ***roll*** and ***marks***as 101 and 56.25, respectively. 4.75

```
#include<stdio.h>
#include<conio.h>

structStudent_struct{
int x;
        float y;
        };

union Student_union{
int x;
        float y;
        };

main()
{
Student_struct s1;
Student_union s2;

printf("Using Stucture \n\n");
printf("Enter the roll of a student\n");
scanf("%d",&s1.x);
printf("\nRoll number is %d and marks is %f",s1.x,s1.y);
```


2023/03/05 11:22

CamScanner

```
ba
cba

void possibleSubsets(char A[], int N)
   {
      for(int i = 0;i < (1 << N); ++i)
      {
         for(int j = 0;j < N;++j)
            if(i & (1 << j))
printf("%c",A[j]);
printf("\n");
   }
   }

int main()
   {
      char A[]={'a','b','c'};
possibleSubsets(A,3);
      return 0;
   }
```

6. (a) What is dynamic memory allocation? **3.00**
   (b) Allocate memory for 100 integer numbers using dynamic memory allocation function in C. Store the numbers into the allocated memory. Your task is to write a program to find the largest number. **5.75**

7. (a) What is the purpose of using a header file? **2.00**
   (b) What is a structure? How does a structure differ from an array? **3.00**
   (c) In the following fragment of code what is the final value of a, b, c. **3.75**
   x=10; b=++x; c=b--; a=--b+a++;

8. (a) What conditions must be satisfied by all the elements of any given array? **2.00**
   (b) A new sorting approach is given. In this approach only one operation (Flip) is available and that is you can exchange two adjacent numbers. If you think a while, you will see that it is always possible to sort a set of numbers in this way. **6.75**

   A set of integers will be given. Now using the above approach you want to sort the numbers in ascending order. You have to find out the minimum number of flips required. Such as to sort '1 2 3'you need no flip operation, however, to sort '2 3 1' you need at least 2 flip operations.

   Input

   The input will start with a positive integer N ($N \leq 1000$). In next few lines there will be N integers. Input will be terminated by 0.

   Output

   For each data set print 'Minimum exchange operations: M' where M is the minimum flip operations required to perform sorting. Use a separate line for each case.

   Sample Input

   3
   1 2 3
   3
   2 3 1
   0

   Sample Output

   Minimum exchange operations : 0
   Minimum exchange operations : 2

2023/03/05 11:21

CamScanner

10-14

University of Rajshahi
(For Affiliated Colleges)
Department of Computer Science and Engineering
B.Sc. Engineering Part-1 Odd Semester Examination 2021
Course: CSE1121 (Structural Programming)
(According to the Syllabus:2019-2020)
Marks: 52.50 Time: 3 Hours
(Answer any THREE questions from each Section)

Section-A

1.(a) What do you know about "if-else-if ladder"? Briefly describe with example. 3.75

(b) (i) If *int i = 5, char c = 'a', int d*, What will be the output of d where 'd= (*i + c)*' ? 2
(ii) If *float num = 25.5*, What will be the output of 'num % 2' and '((int) num) % 2'?

(c) Write a C program to find the largest from three numbers given to explain 'Nested if'. 3

2.(a) What is run-time error? Give some examples of run-time error? 1.75

(b) What will be output of the following code segment? 3.5

```
int main()
{
   int a[]= {1,2,3,4,5,6,7,8,9,0};
   p(&a[4]);
   return 0;
}
```

```
void p(int *b)
{
 int i,s=0;
 for (i=0; i<5;i++)
 {
  s= s+ *b;
  b--;
 }
 printf("%d",s);
}
```

(c What will be output of the following code segment? 3.5

```
int main()
{
   int a[]= {1,2,3,4,5,6,7,8,9,0};
    p(&a[4]);
   return 0;
}
```

```
void p(int b[])
{
int i=0,s=0;

 while (b[i])
 {
  s= s+ b[i];
  i++;
 }
 printf("%d",s);
}
```

3. (a) Explain the 'break' and 'continue' statements with example. 3

(b) How do-while and while statements differ? Breifly discuss with example. 3

(c) Find the output of the following program code with explanations. 2.75

```
#include<stdio.h>
void main()
{
   int i, j, x=0;
   for(i=1; i<4; i++)
 { for(j=1; j<=3; j++)
    { if(i==j) continue;
        x=x+i+j;
  }
     printf("\nx=%d\n",x);
  }
 }
```

2023/03/05 11:20

CamScanner

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-I OddSemester, Examination-2017
Course: CSE1121(Computer Programming with C)
Time: 3 Hours Total Marks 52.5

*(Answer any **SIX (06)** questions taking **THREE (03)** from each section)*

Section A

1. (a) If *int i = 5, float f = 7.5, char c = 'a'*, what will be the output of (i) ***i + c*** and (ii) ***i + f***? 1
   (b) If i = 10, f = 18.5 and *int result*, what will be the output of ***result = (i + f) % 4***? 1
   (c) If *float num = 25.5*, what will be the output of***num % 2***and ***((int)num) % 2***? 1
   (d) List the syntax error (if any) of each line of the following code (copy the code on your script, mark the error by placing an underline and then explain that error). 5.75

```
#include<stdio>
main()
{
Intx,y;
scanf("%dd",&x,&y);
switch (x)
   {
    case(x<0): printf("x is positive"); break;
    case(x>0);printf("x is negative");
   }
x=f1(y);
printf("Final value of = %d\n", x);
}

void f1(int y)
{
y=y*y;
return y;
}
```

2. (a) Explain 'the name of array can be considered as pointer'. Give an example how you can pass array to a function. 3.75
   (b) Write a C program that will take N number characters as input and a query character Q. You have to find out how many times Q occurs in that input character array. 5

Input: The input will have one integer N (100>N>0), then N number of characters and finally a query character Q.
For example:
20
a g b h b r a d f v c d b a f a g b a a
a

Output: The output will be the number of how many times Q occurs in that input array.
For example; the output for the above input will be: 6

3. (a) What do you mean by hierarchy of operation? Determine the hierarchy of operations and evaluate the following expression: 3.75
   $i = 2*3/4+4/4+8-2+5/8.$

2023/05/05 11:22
CamScanner

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering **Part-I, Odd** Semester Examination **2018(2016-2017)**
Course Code: **CSE-1121** Course Title: **Computer Programming with C**
Time: **03** Hours Full Marks: **52.5**

Section **A**
Answer any **THREE** questions.

1. (a) What is run-time error? Give some examples of run-time error? **2.25**
   (b) If i=4, j=2.5 and int answer, what will be the output of answer=(i+j)%2? **1.50**
   (c) List the syntax error (if any) of each line of the following code. **5.00**

```
struct Books {
  char  title[50];
  char  author[50];
  char  subject[100];
  int   book_id;
};
double printBook( struct Books book ) {
   printf( "Book title : %f\n", book.title);
   printf( "Book author : %s\n", book.author);
   printf( "Book subject : %s\n", book1.subject);
   printf( "Book book_id : %s\n", book.book_id);
}

int main( ) {

   struct Books Book1;
   struct Books Book2;

   strcpy( Book1.title, "C Programming");
   strcpy( Book1--author, "Prata");
   strcpy( Book1.subject, "C Programming Tutorial");
   Book1.book_id = '6495407';

   strcpy( "Telecom Billing", Book2);
   strcpy( Book2.author, "Goodman");
   strcpy( Book2.subject, "Telecom Billing Tutorial");
   Book2.book_id = "6495700";

   printBook( Books Book1 );

   printBook( Book2 );

   return 1;
}
```

2. (a) What will be the output of the following code? **3.75**

```
#include <stdio.h>

int a = 20;
int e;

int main () {
  int a = 10;
  int b = 20;
```

2023/03/05 11:22

CamScanner

(b) Convert the equation into corresponding C statements: $R = \frac{2v + 6.22(c+d)}{g+v}$.

(c) What would be the output of the following program:

```
main()
{
int i=2, j=3,k,l;
   float a,b;
   k=i/j*j; l=j/i*i;
   a=i/j*j; b=j/i*i;
printf("%d%d%f%f", k,l,a,b);
}
```

4. (a) What are the differences between compiler and interpreter? 2

(b) A student from the dept. of CSE developed the following code to get the expected output as given below. However, unfortunately his code has a single bug; therefore his code did not produce the expected output. 6.75

(i) What is the current output?

(ii) What is that single bug?

(iii) Fix that single bug to get the expected output (you can change/modify only one operator or replace only one variable by another one or change/replace value assigned to only one variable)

```
#include<stdio.h>
#include<conio.h>

inti,j,k,x,y,z=3;

main()
{ for (i=1;i<=z;i++)
 {  x=i;
for(j=1; j<=z ; j++)
{  y=j;
if(x!=y)
        {for (k=65; k<65+z; k++)
printf("%c ",k);
        }
else
        {for (k=64+z; k>=65; k--)
printf("%c ",k);
        }
}
printf ("\n");
}
getch();
}
```

Expected output:

A B C C B A C B A
C B A A B C C B A
C B A C B A A B C

2023/03/05 11:22

CamScanner

Section-B

5. (a) What will be output of the following program?

```
#include<stdio.h>
#include<conio.h>

int x[5][5]={  {1, 2, 3, 4, 5},
               {6, 7, 8, 9, 7},
               {0, 1, 2, 3, 4},
               {5, 6, 2,8, 9},
               {8, 7, 6, 5, 4} };

int i, j, k, l, tmp, big, p;

main() {

 for (i=0;i<=4;i++)
 {
   for(j=i; j<=4; j++)
   {
     for(k=1; k<=4; k++)
     {
       for(l=k ; l<=4; l++)
       {
       x[k][l]=x[k][l]+1;
       }
     }

   }
  }
 for (i=0;i<=4;i++)
  {
    for(j=0; j<=4; j++)
    {
printf ("%d ",x[i][j]);
    }
printf ("\n ");
  }

getch();
}
```

(b) 100 numbers are written in a file named input.txt. In each line, 10 numbers are written. All the numbers are separated by a single space. Write a program to determine the frequency of the numbers and write into another file named output.txt. 5.00

6. (a) Given an array o... in the given array... equals the given in...

(b) Given an...

2023/03/05 11:21

CamScanner

```
int main()
{
int b;
   c=100;
   a=40;
printf("\n Before calling a=%d b=%d c=%d",a,b,c);
addint(a);
printf("\n After calling a=%d b=%d c=%d",a,b,c);
   return 0;
}
```

4. (a) Write a program to take three matrix A[n][n],B[n][n] and C[n][n]from the keyboard. Set the value of each cell of a row of the matrix D[n][n] with the largest value of respective three rows of matrix A, B and C.The maximum size of 'n' is 10. 4.50

Sample input:

Value of n=3

A			B			C		
1	2	3	2	3	4	1	1	2
4	5	6	1	2	3	2	3	4
2	4	5	7	8	9	6	1	2

Sample output:

D		
4	4	4
6	6	6
9	9	9

(b) Let a[5][5]={1 2 3 4 5 4.25
2 3 4 5 6
3 4 5 6 7
4 5 6 7 8
5 6 7 8 9 }

What will be output of the following code segment?

```
for (i=2; i<=4; i++)
{
    p=i-1;
    for (j=1; j<=4; j++)
    {if (i!=j) p=p+a[i][j]+1; else p=p+a[i][j]-1;}
    printf("%d\n"p);
}
```


2023/03/05 11:21

CamScanner

4. (a) What will be the output of the following code?

```
#include <stdio.h>
int a = 20;
int sum(int a, int b)
{
printf ("value of a in sum() = %d\n",  a);
printf ("value of b in sum() = %d\n",  b);
  return a + b;
}
```

```
int main () {
int a = 10;
int b = 20;
int c = 0;
printf ("value of a in main() = %d\n",  a);
 c = sum( a, b);
printf ("value of c in main() = %d\n",  c);
 return 0;
}
```

4.25

(b) What will be the output of the following code?

```
#include<stdio.h>
int a,b,c;
void addint(int a)
{
   a=50;
   b=200;
   c=300;
}
```

```
int main(){
int b;
   c=100;
   a=40;
printf("\n Before calling a=%d b=%d
c=%d",a,b,c);
addint(a);
printf("\n After calling a=%d b=%d
c=%d",a,b,c);
   return 0;
}
```

Section-B

5. (a) What is pointer? What are null pointer and void pointer? 2.75

(b) How are pointers and arrays closely related? Explain with example. 3

(c) What is the output of the following code? If there is any error, then rewrite the code by correcting the error and give the output with explanation. 3

```
#include<stdio.h>
  int main()
  {
     int *p, x;
    *p = 2;
     x = 15 /*p;
     printf("%d", x);
  }
```

6. (a) What is structure? Explain with a suitable example, how are the members of a structure accessed? 3

(b) Write the differences between structure and union. 2

(c) Write a C program to define structure for class containing *class name*, *no. of students* and *block no.* Input 5 records from keyboard and display it. 3.75

7. (a) Describe different types of file opening mode to open a data file. 4

(b) Write a C program that will read a text file and count the total number of vowels within the text file (the file only contains uppercase letters). 4.75

8. (a) What will be output of the following code segments? 3.5

(i)

```
int i = 0;

while (i < 10) {
 if (i == 4) {
   break;
 }
 printf("%d\n", i);
 i++;
}
```

(ii)

```
int i = 0;

while (i < 10) {
 if (i == 4) {
   i++;
   continue;
 }
 printf("%d\n", i);
 i++;
}
```

(b) Write a C program to print all prime numbers among 1 to n. 3.5

(c) What is debugging? 1.75

2023/03/05 11:20

CamScanner

```
    tempnum = *num1;
    *num1 = *num2;
    *num2 = tempnum;
}
int main( )
{
    int v1 = 11, v2 = 77 ;
    printf("Before operation:");
    printf("\nValue of v1 is: %d", v1);
    printf("\nValue of v2 is: %d", v2);

    operation( &v1, &v2 );

    printf("\nAfter operation:");
    printf("\nValue of v1 is: %d", v1);
    printf("\nValue of v2 is: %d", v2);
}
```

(b) A 4x4 matrix is given in a file named in.txt. **5.75**

in.txt:

10	15	20	25
5	10	20	15
15	25	10	5
20	5	15	10

Read the matrix from the in.txt file, transform all the decimal numbers to its equivalent binary numbers and write into out.txt file.

out.txt:

1010	1111	10100	11001
101	1010	10100	1111
1111	11001	1010	101
10100	101	1111	1010

Section **B**

Answer any **THREE** questions.

5. (a) Some numbers are given as input and their average 214.400000 is calculated using getAverage() function. However, getAverage() is not defined. Your task is to define getAverage() function. **3.00**

```
int main () {
  int balance[5] = {1000, 2, 3, 17, 50};
  double avg;
  avg = getAverage( balance, 5 ) ;
  printf( "Average value is: %f ", avg );
  return 0;
}
```

(b) A programmer developed the following program to have the expected output as given below. However, expected output is not found. **5.75**

(i) What is the current output?
(ii) What is the error?
(iii) Make necessary correction.

Expected Output:

```
c
b
cb
a
ca
```

2023/03/05 11:22

CamScanner

University of Rajshahi
(Affiliated College)
Department of Computer Science and Engineering

B. Sc. (Engg.) Part-I Odd Semester Examination 2021
Course: CSE 1121 (Structural Programming Language)
Full Marks: 60 Duration: 3 (Three) Hours

Answer 06 (Six) questions taking any 03 (Three) questions from each part

Section-A

1. a) Define identifier. Determine which of the following are valid identifiers? If invalid, explain why. 3

   i) $tax ii) name iii) 1record
   iv) file_3 v) extern vi) R_university

   b) In the following fragment of code, what is the final value of a, b, c? 3

   i) x = 10; b = ++x; c = b--; a = --b + c++;
   ii) x = 2; a = 1; c = 3; b = ++a * x - c--;

   c) Write a sample 'c' program that takes three numbers from keyboard and find the maximum number among them. 4

2. a) Define operators. Describe two different ways to use the unary increment and decrement operators. 4.5

   b) Explain the meaning of each of the following function declaration. 3.5

   i) void func (int a, double b).
   ii) double func (int a, int b).
   iii) char func (void).

   c) What will be the output of the following program when executed? 2

```
void main () {
int i, j, x =5, y=6;
for i=1; i<=5; i++) {
    printf ("%d\n", x++);
    if (x>=8) break; }
for i=1; i<=5; i++) {
    if (y>=10) break;
    printf ("%d\n", ++y); }}
```

3. a) What is called looping? What are the entry controlled and exit controlled loop? Explain with proper example. 3.5

   b) Write a 'c' program using for or while loop to create the output as follows: 4

```
                1
              2 1 2
            3 2 1 2 3
          4 3 2 1 2 3 4
        5 4 3 2 1 2 3 4 5
```

   c) Describe the difference between nested if-else and switch statement using suitable examples. 2.5

4. a) What is an array variable? What are the benefits of using array variable over ordinary variable? 3

   b) Write a 'c' program that read two (3×3) matrices. Calculate their multiplication and store the result on a third array variable. 4.5

   c) What will be the output of the following variables when the code executed? 2.5

   i)
```
int a = 5, b = 6;
int x = ++a - b;
printf ("a = %d, x = %d", a, x);
```

   ii)
```
int a = 5, b = 6;
int y = a++ - b++;
printf ("b = %d, y = %d", b, y);
```

2023/03/05 11:20

CamScanner

ba
cba

```
void possibleSubsets(char A[], int N)
{
   for(int i = 0;i < (1 << N); ++i)
   {
      for(int j = 0;j < N;++j)
         if(i & (1 << j))
            printf("%c",A[j]);
      printf("\n");
   }
}

int main()
{
   char A[]={'a','b','c'};
   possibleSubsets(A,3);
   return 0;
}
```

6. (a) What is dynamic memory allocation? **3.00**
   (b) Allocate memory for 100 integer numbers using dynamic memory allocation function in C. Store the numbers into the allocated memory. Your task is to write a program to find the largest number. **5.75**

7. (a) What is the purpose of using a header file? **2.00**
   (b) What is a structure? How does a structure differ from an array? **3.00**
   (c) In the following fragment of code what is the final value of a, b, c. **3.75**
   x=10; b=++x; c=b--; a=--b+a++;

8. (a) What conditions must be satisfied by all the elements of any given array? **2.00**
   (b) A new sorting approach is given. In this approach only one operation (Flip) is available and that is you can exchange two adjacent numbers. If you think a while, you will see that it is always possible to sort a set of numbers in this way. **6.75**
   A set of integers will be given. Now using the above approach you want to sort the numbers in ascending order. You have to find out the minimum number of flips required. Such as to sort '1 2 3'you need no flip operation, however, to sort '2 3 1' you need at least 2 flip operations.

   Input
   The input will start with a positive integer N ($N \leq 1000$). In next few lines there will be N integers. Input will be terminated by 0.

   Output
   For each data set print 'Minimum exchange operations: M' where M is the minimum flip operations required to perform sorting. Use a separate line for each case.

   Sample Input
   3
   1 2 3
   3
   2 3 1
   0

   Sample Output
   Minimum exchange operations : 0
   Minimum exchange operations : 2

2023/03/05 11:22

CamScanner